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(54) **ANTIBODY OR ANTIBODY FRAGMENT THAT SPECIFICALLY BINDS TO VOLTAGE-GATED SODIUM CHANNEL ALPHA SUBUNIT NAV1.7**

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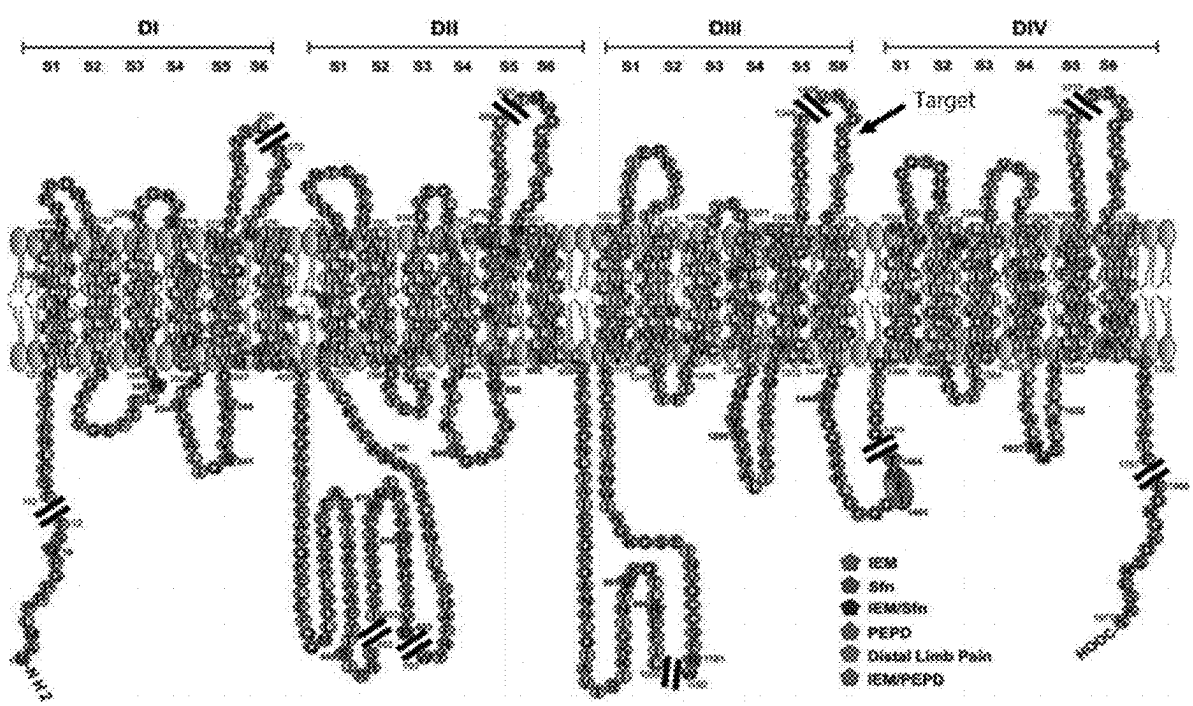
(57) **ABSTRACT**

The present invention provides an antibody or antibody fragment that targets cell membrane voltage-gated sodium ion channel α subunit Nav 1.7; a specific binding target thereof is an ion-conducting pore module (PM) of an S3 domain of domain IV of a voltage-gated sodium ion channel α subunit. The antibody or antibody fragment thereof can inactivate the ion-conducting PM, so that sodium ions cannot normally enter nerve cells, to thereby achieve the effect of treating and relieving pain.

Specification includes a Sequence Listing.



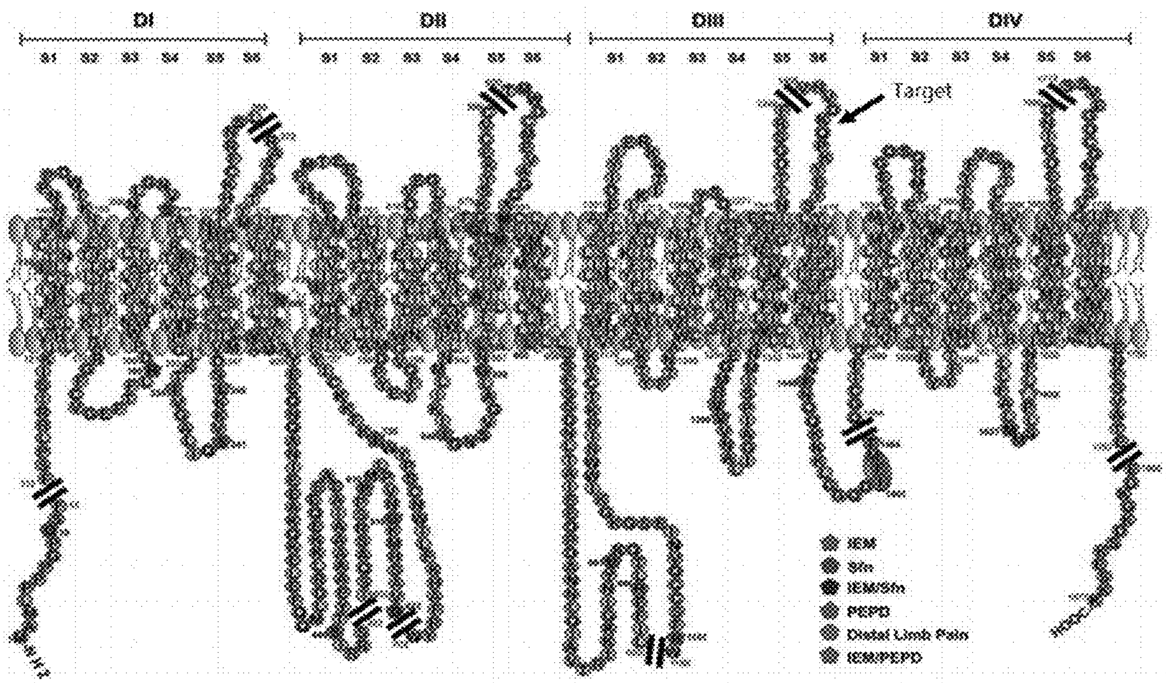

DI
DII
DIII
DIV
S1 S2 S3 S4 S5 S6
Target
IEM
Sfn
IEM/Sfn
PEPD
Distal Limb Pain
IEM/PEPD

Fig. 1

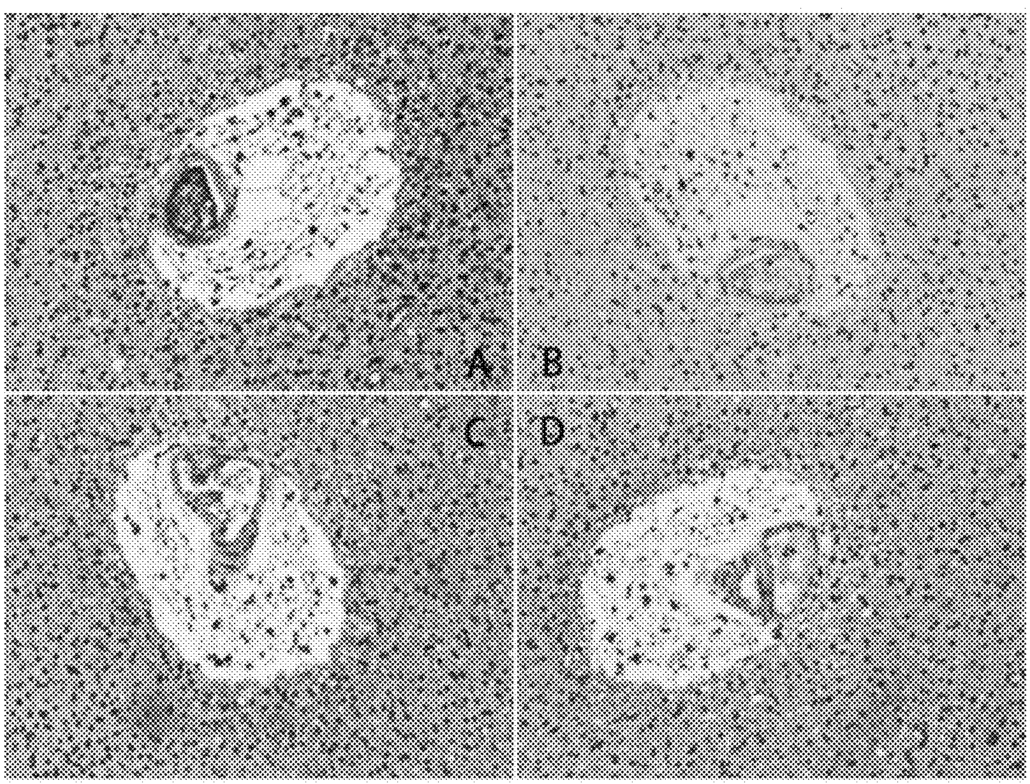

A
B
C
D

Fig. 2

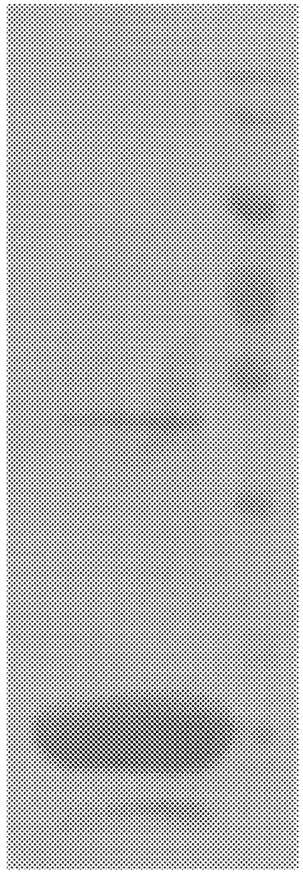
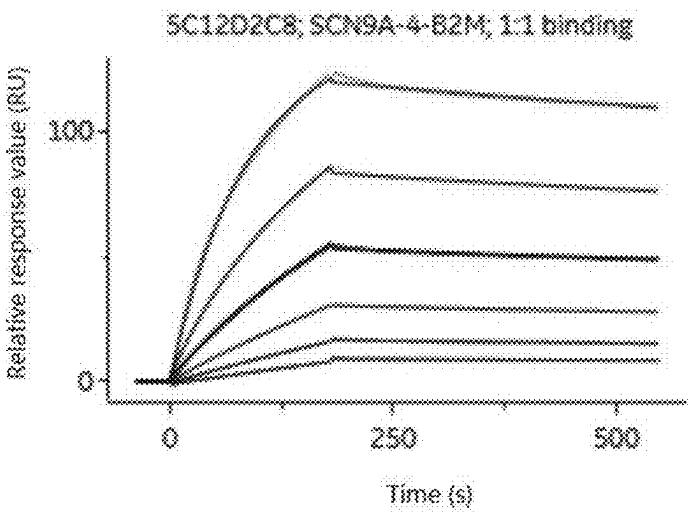


Fig. 3



Ligand	Analyte	k_a (1/Ms)	k_d (1/s)	K_D (M)	Chi² (RU²)
SCN9A-4-B2M	5C12D2C8	2.81E+04	2.47E-04	8.78E-09	3.79E-01

Fig. 4

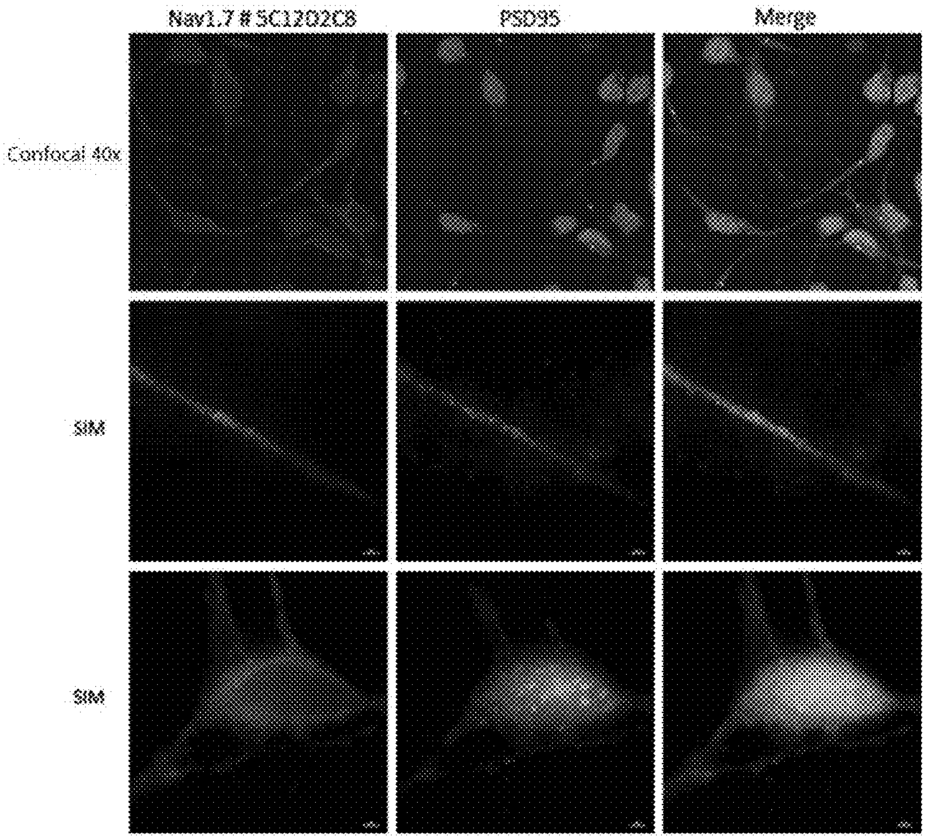

Nav1.7 # SC12D2C8
PSD95
Merge
Confocal 40x
SIM
SIM

Fig.5

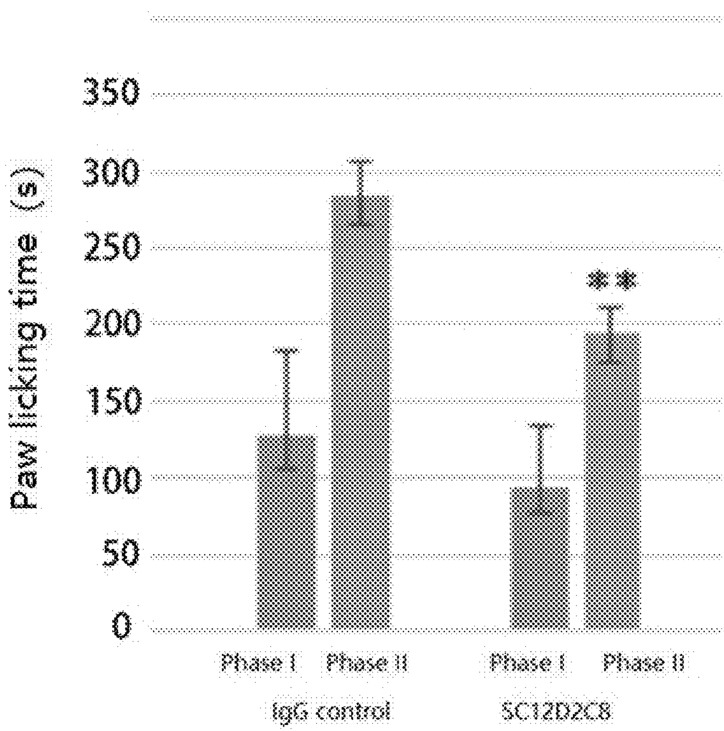

Paw licking time (s)
350
300
250
200
150
100
50
0
**
Phase I
Phase II
Phase I
Phase II
IgG control
SC12D2C8

Fig. 6

ANTIBODY OR ANTIBODY FRAGMENT THAT SPECIFICALLY BINDS TO VOLTAGE-GATED SODIUM CHANNEL ALPHA SUBUNIT NAV1.7

FIELD OF THE INVENTION

[0001] The present invention belongs to the field of biomedicine and relates to an antibody and/or antibody fragment that targets the ion-conducting pore module of Nav1.7 and specifically recognizes the target (polypeptide).

REFERENCE TO SEQUENCE LISTING SUBMITTED VIA EFS-WEB

[0002] This application includes an electronically submitted sequence listing in .txt format. The .txt file contains a sequence listing entitled "WH1917-23P151186US ST25.txt" created on Nov. 16, 2023 and is 15,306 bytes in size. The sequence listing contained 20 in this .txt file is part of the specification and is hereby incorporated by reference herein in its entirety.

BACKGROUND OF THE INVENTION

[0003] Pain begins with the nociceptors of the peripheral nerve system, and peripheral nerve tissues are widely distributed in the skin, muscles, joints and visceral tissues of the whole body as a kind of free nerve ending, and can convert thermal, mechanical or chemical stimuli into action potentials, transmit them to the cell body in the dorsal root ganglia (DRG) through nerve fibers and ultimately to the advanced nerve center, thereby causing pain. The generation and conduction of action potentials in neurons in turn depend on the voltage-gated sodium channels (VGSCs) located on the cytomembrane. When the cytomembrane is depolarized, the sodium ion channel is activated. The channel is opened, causing sodium ion influx, and further depolarizing the cytomembrane, resulting in the generation of an action potential, and thus causing pain due to the abnormal action potential. Therefore, inhibition of abnormal sodium ion channel activity contributes to the treatment and alleviation of pain.

[0004] Voltage-gated sodium channels can be categorized into nine subtypes. At present, nine a-subtypes of voltage-gated sodium channels have been identified in mammals, since their amino acid sequences have more than 50% similarity, they are considered to be from the same family, named Nav1 (Nav1.1-Nav1.9). Voltage-gated sodium channels, widely found on the cytomembranes of neuron and skeletal muscle cells, are a class of transmembrane glycoprotein complexes composed of an α subunit and a number of auxiliary β subunits. The α-subunit is composed of two functional domains, i.e. an ion-conducting pore domain and voltage-sensing domains (VSDs). The pore of the α-subunit consists of four repeating domains (DI-DIV), each of which contains six transmembrane helical fragments (S1-S6). S1-S4 contain the voltage-sensing domain VSD, and S5-S6 form a tetrameric conformation to form the pore domain. Within the VSD domain, S4 contains the VSD, and S4 is enriched in gating-charge arginine, which senses changes in membrane potential. Together with the C-terminus of S3, it forms a voltage-sensor paddle, whose movement reflects changes in membrane potential and is coupled to pore opening, closing, and inactivation. Since this voltage-sensor paddle movement is the opening and locking of the channel. Therefore, this functional structural domain is an important drug action target and can be used through protein interactions to regulate the switching of channels.

[0005] Recent studies have shown that the main subtypes of Nav1 associated with pain are Nav1.7, Nav1.8, and Nav1.9. Among them, Nav1.7 is one of the important members mainly responsible for pain. Nav1.7 is a TTX-S type and encodes the gene SCN9 A. It is mainly distributed in peripheral primary sensory neurons and sympathetic ganglion neurons, and is involved in the human pain signaling pathway. Recently, in human pain-free patients, pain-free symptoms have occurred following genetic mutations in Nav1.7; further studies have shown that this gene is one of the sodium ion channels primarily responsible for pain.

[0006] Small chemical molecules (e.g., carbamazepine, lidocaine, and mexiletine) are commonly used clinically as voltage-gated sodium channel inhibitors for the treatment of pain; however, they lack sufficient selectivity for voltage-gated sodium channel subtypes, and thus have the disadvantage of cardiotoxicity and central nervous system side effects. Recently, some small-molecule blockers against Nav1.7 have been introduced into clinical studies, but due to the high homology of sodium channel subtypes, small-molecule blockers have poor selectivity and their side effects are difficult to overcome. Large molecule blockers have high specificity, good stability, and fewer side effects, but they are difficult to study because the antibody-producing voltage-gated sodium channel antigen is not easy to prepare.

SUMMARY OF THE INVENTION

[0007] An object of the present invention is to provide an antibody or an antibody fragment that specifically binds to the voltage-gated sodium channel α-subunit Nav1.7. The specific binding means that the ion-conducting pore module (PM) of the DIVS3 domain of Nav1.7 is used as a target, the target in this region is used to design a polypeptide as an antigen to obtain a monoclonal antibody, and the specific antibody can interfere with the normal state of the VGSCs through binding to its target, thereby inhibiting pain.

[0008] A second object of the present invention is to provide a pharmaceutical composition comprising the antibody or antibody fragment thereof that specifically binds to the ion-conducting pore module of the DIVS3 domain of the voltage-gated sodium channel α-subunit Nav1.7.

[0009] A third object of the present invention is to provide uses of the antibody or antibody fragment thereof that specifically binds to the ion-conducting pore module of the DIVS3 domain of voltage-gated sodium channel α-subunit Nav1.7, or of the pharmaceutical composition.

[0010] The present invention also provides a nucleotide encoding the antibody or antibody fragment, an expression vector containing the nucleotide, and a method of preparing the antibody or antibody fragment.

[0011] According to an aspect of the present invention, an antibody or an antibody fragment thereof that specifically binds to the ion-conducting pore module of the DIVS3 domain of the voltage-gated sodium channel α-subunit Nav1.7 of the present invention, the target of specific binding is the ion-conducting pore module of the DIV/S3 domain of the voltage-gated sodium channel α-subunit. More preferably, the amino acid sequence of the bound antigen is: DSVNVDKQPKYEYS (SEQ ID NO.9).

[0012] Based on the crystal structure of Nav1.7, the ion-conducting pore module of the DIV/S3 domain of the voltage sensor valve of Nav1.7 was used to screen suitable polypeptides in the target region as an antigen, and after the hydrophilicity and antigenicity analysis, a polypeptide with good hydrophilicity and high antigenicity was selected, having the amino acid sequence of DSVNVDKQPKYEYS (SEQ ID NO.9).

[0013] The above polypeptide was chemically synthesized, the synthesized polypeptide was numbered C9797BL020-7 (SEQ ID NO. 9), coupled to the carrier protein KLH, and then immunized to BALB/c mice. The body was stimulated by multiple immunizations to produce an immune response, which resulted in a polyclonal antibody for blood collection, ELISA assay and evaluation.

[0014] Based on the antigen-antibody reaction, the potency of the polyclonal antibody produced by the immunized animals was evaluated by ELISA. Based on the antibody potency of the immunized animals and the specificity of the human neural tissues, three animals that met the requirements, #4061, #4062 and #4063, were finally identified for cell fusion. The spleen cells of the three animals were electrofused with mouse myeloma cells (SP2/0) and then subjected to cell culture after fusion. The positive cell lines were screened on screening medium, and hybridoma cell lines were screened by using the polypeptide C9797BL020-7 as an antigen. Based on the ELISA results, positive cell lines were selected for subcloning based on antibody potency and human neural tissue specificity. The resulting subclones were again subjected to ELISA and neural tissue specificity assays, and those with good specificity for nerve tissues were selected for subcloning for cell cryopreservation.

[0015] Total RNA of the cell line was extracted, cDNA was synthesized, a cDNA library was created, and variable regions were sequenced. The polynucleotide sequences encoding the variable regions of the antibody were amplified, and the DNA sequences encoding VH and VL (which could also be manipulated with RNA sequences encoding the variable regions) could be integrated into the same vector, or they could be integrated into the vector separately, and suitable host cells were transfected with the above vector; and then sequenced and analyzed. The sequencing results showed that the VH has the DNA sequence as shown in SEQ ID NO: 10 and the VL has the DNA sequence as shown in SEQ ID NO: 11.

[0016] Genetically engineered antibodies were constructed. According to different needs, the above DNA sequences encoding VH and VL (or encoding CDR in VH and encoding CDR in VL) were introduced into a suitable host for antibody expression, and the antibody effect was verified.

[0017] The immunogenicity of the monoclonal antibody was tested. The target sequence was prokaryotically expressed, and the total protein was extracted from the prokaryotically expressed bacteria, and then subjected to preliminary purification to obtain an antigenic fragment. The binding specificity of the antibody was analyzed by Western Blotting, as shown in FIG. **4**, the antibody can specifically recognize the target protein sequence of Nav1.7. An acute inflammatory pain mice model was established by treating with 5% formalin. An appropriate amount of antibody was injected via tail vein to detect the analgesic effect of the antibody on the pain model mice. The results as shown in FIG. **5**, showed that the injection of 10 mg/kg of antibody had a significant analgesic effect over the control.

[0018] According to another aspect of the present invention, an antibody or an antibody fragment thereof that specifically binds to the voltage-gated sodium channel α subunit Nav1.7, comprises:

- **[0019]** heavy chain complementary determining regions HCDR1, HCDR2, HCDR3, the HCDR1 having the amino acid sequence as shown in SEQ ID NO. 1, the HCDR2 having the amino acid sequence as shown in SEQ ID NO. 2, the HCDR3 having the amino acid sequence as shown in SEQ ID NO. 3; and
- **[0020]** light chain complementary determining regions LCDR1, LCDR2, LCDR3, the LCDR1 having the amino acid sequence as shown in SEQ ID NO. 4, the LCDR2having the amino acid sequence as shown in SEQ ID NO. 5, the LCDR3 having the amino acid sequence as shown in SEQ ID NO. 6.

[0021] According to the present invention, the antibody is a monoclonal antibody or a polyclonal antibody. Preferably, the antibody is a monoclonal antibody.

[0022] According to the present invention, the antibody is a murine antibody, a chimeric antibody or a humanized antibody, etc. Preferably, the antibody is a humanized antibody.

[0023] According to the present invention, the antibody fragment comprises the forms of Fab, F(ab')2, dsFv, scFv, a diabody, a minibody, a bispecific antibody, a multi-specific antibody, a chimeric antibody, and a CDR-grafted antibody.

[0024] According to a preferred embodiment of the present invention, the antibody or antibody fragment that specifically binds to the voltage-gated sodium channel α subunit Nav1.7 comprises a heavy chain variable region and a light chain variable region. The heavy chain variable region has the amino acid sequence as shown in SEQ ID NO. 7, and the light chain variable region has the amino acid sequence as shown in SEQ ID NO. 8. It will be appreciated by those skilled in the art that the antibody or antibody fragment of the present invention also comprise a structurally similar derivative sequence with 80%, 80-85%, 85-90%, 90-95% or 95-99% homology to the sequence of SEQ ID NO: 7/SEQ ID NO.8 of the heavy/light chain variable region.

[0025] According to a preferred embodiment of the present invention, the antibody or antibody fragment that specifically binds to the voltage-gated sodium channel α subunit Nav1.7 comprises a heavy chain constant region and a light chain constant region. The species sources of the light chain constant region and the heavy chain constant region may be selected from the group consisting of human antibody constant region, bovine antibody constant region, sheep antibody constant region, canine antibody constant region, porcine antibody constant region, feline antibody constant region, equine antibody constant region, and donkey antibody constant region. Preferably, the heavy chain constant region is selected from the group consisting of IgG1, IgG2, IgG3 and IgG4 heavy chain constant regions and the light chain constant region is selected from the group consisting of κ or λ light chain constant region. Preferably, the heavy chain constant region is an IgG4 heavy chain constant region and the light chain constant region is a κ light chain constant region.

[0026] According to a preferred embodiment of the present invention, the antibody or antibody fragment that specifically binds to the voltage-gated sodium channel α subunit Nav1.7 comprises a heavy chain and a light chain. The heavy chain has the amino acid sequence of SEQ ID NO: 14 and the light chain has the amino acid sequence of SEQ ID NO: 15.

[0027] According to a further aspect of the present invention, provided is a nucleotide sequence. The nucleotide sequence encodes the antibody or antibody fragment that specifically binds to the voltage-gated sodium channel α subunit Nav1.7 of the present invention.

[0028] According to a preferred embodiment of the present invention, the nucleotide sequence comprises: a nucleotide sequence encoding the heavy chain variable region as shown in SEQ ID NO: 10, a nucleotide sequence encoding the light chain variable region as shown in SEQ ID NO: 11.

[0029] According to a preferred embodiment of the present invention, the nucleotide sequence comprises: a nucleotide sequence encoding the heavy chain as shown in SEQ ID NO: 16, a nucleotide sequence encoding the light chain as shown in SEQ ID NO: 17.

[0030] The present invention also provides an expression vector, which contains anyone of the nucleotide sequence described above.

[0031] The present invention also provides a host cell, which contains the expression vector described above.

[0032] The present invention also provides a method of preparing the antibody or antibody fragment that specifically binds to the voltage-gated sodium channel α subunit Nav1.7 described above, the method comprises the following steps of:

- [0033] (a) culturing the host cell described above under expression conditions so as to express the antibody or antibody fragment that specifically binds to the voltage-gated sodium channel α subunit Nav1.7;
- [0034] (b) isolating and purifying the antibody or antibody fragment that specifically binds to the voltage-gated sodium channel α subunit Nav1.7 of (a).

[0035] According to another aspect of the present invention, provided is a pharmaceutical composition. The pharmaceutical composition comprises the antibody or antibody fragment that specifically binds to the voltage-gated sodium channel α subunit Nav1.7 described herein as an active ingredient, and a pharmaceutically acceptable carrier. The pharmaceutical composition has an analgesic and pain threshold raising effect and can treat pain, itching and cough.

[0036] According to a further aspect of the present invention, provided is the use of the antibody or antibody fragment that specifically binds to the voltage-gated sodium channel α subunit Nav1.7 described herein or a pharmaceutical composition described herein in the manufacture of a medicament for the treatment of pain-related diseases.

[0037] Beneficial effects: The specific binding of the antibody, a biomolecule targeted against the voltage sensor of the Nav1.7 voltage-gated sodium channel, is used to inactivate the ion-conducting pore module of the DIVS3 domain, so that sodium ions cannot normally enter nerve cells, thus achieving the effect of treating and relieving pain. Due to its excellent targeting properties, it can overcome the side effects due to chemical small molecule drugs.

DESCRIPTION OF THE DRAWINGS

[0038] FIG. **1**: Structural diagram of the sodium channel Nav1.7 and schematic of the target;

[0039] FIG. **2**: Immunohistochemical analysis of human neural tissue incubated with sera from immunized animals;

[0040] A. 4061; B. 4062; C.4063; D. 4064; E. 4065

[0041] FIG. **3**: Western blotting immunogenicity analysis of monoclonal antibodies;

[0042] FIG. **4**: Affinity binding curve of monoclonal antibody 5C12D2C8 to Nav1.7 target determined by SPR;

[0043] FIG. **5**: Localization of monoclonal antibody 5C12D2C8 that targets to Nav1.7 in human peripheral nerve cells;

[0044] FIG. **6**: Analgesic effect of 5C12D2C8 antibody on 5% formalin-induced acute inflammatory pain in wild-type mice.

DETAILED DESCRIPTION OF THE INVENTION

[0045] The invention is illustrated, but not limited by the following detailed description of the preferred embodiments of the invention.

Material Sources

[0046] The materials and reagents used below are commercially available unless otherwise stated.

Example 1 Synthesis of Antigen

[0047] According to the amino acid sequence (GenBank No. NP_002968) and the functional regions of the crystal structure of Nav1.7 (FIG. **1**), hydrophilicity and antigenicity analysis was performed. The sequence of DSVNVDKQPKYEYS was screened, the hydrophilicity and antigenicity of which met the requirements of the antigen. The CDSVNVDKQPKYEYS (SEQ ID NO.9) polypeptide was synthesized using a fully automated synthesizer.

[0048] Specific steps were as follows:

- [0049] (1) attaching —COOH of the first AA to Cl-Resin with DIEA, and then blocking the unreacted functional groups on the resin with MeOH;
- [0050] (2) washing with DMF;
- [0051] (3) removing the protecting group Fmoc of —NH_2 in the first AA with Pip to expose the —NH_2;
- [0052] (4) washing with DMF;
- [0053] (5) activating —COOH of the second AA with DIC+HOBT, and then condensing it with —NH_2 in the first AA to form an amide bond;
- [0054] (6) washing with DMF;
- [0055] (7) removing the protecting group Fmoc of —NH_2 in the second AA with Pip to expose the —NH_2;
- [0056] (8) washing with DMF;
- [0057] (9) . . . repeating the steps 5-8 until exposing the —NH_2 of the last AA;
- [0058] (10) cutting the polypeptide from the resin and removing the side chain protecting groups of all amino acids with the cleavage reagent as: trifluoroacetic acid+ethanedithiol+phenol+thioanisole+water;
- [0059] (11) adding the cleavage solution into diethyl ether to precipitate the polypeptide, and centrifuging to obtain the crude polypeptide (C9797BL020-7);
- [0060] (12) purifying the polypeptide with the HPLC C18 preparative/analytical column, designated as C9797BL020-7, to obtain the purified polypeptide for immunizing animals.

[0061] Note: To facilitate polypeptide coupling, an additional cysteine may be added to the end of this polypeptide.

Example 2 Preparation of Monoclonal Cell Lines

2.1 Animal Immunization

[0062] Freund's complete adjuvant (Sigma, F5881) and Freund's incomplete adjuvant (Sigma, F5506) were prepared. The polypeptide was coupled to the carrier protein KLH by the terminal SH of polypeptide C9797BL020-7, as an immunogen.

[0063] Five 8-week-old female BALB/c (animal numbers: #4061, #4062, #4063, #4064, #4065) were selected and immunized intraperitoneally three times to stimulate the body to produce an immune response and then to produce antibodies. Primary immunization: 50 μg/each; the secondary immunization was performed after three weeks, at a dose of 50 μg/each; the third immunization was performed 2 weeks after the second immunization at a dose of 50 μg/each; 1 week after the third immunization, blood was collected for antibody test.

2.2 ELISA Test of Animal Serum

2.2.1 Instruments and Equipments

[0064] Washing machine: Beijing Nanhua ZDMX

[0065] Microtiter-plate reader: Thermo Multiskan Ascent

2.2.2 Reagents

[0066] Coating antigen: polypeptide C9797BL020-7; coating solution: 1*PBS (pH 7.4); washing buffer: 1*PBS (pH 7.4), 0.05% PBS; the primary antibody: anti-serum after the third immunization; enzyme-labeled secondary antibody: Peroxidase-AffiniPure Goat Anti-Mouse IgG, Fcγ Fragment Specific (min X Hu, Bov, HrsSrProt); TMB chromogenic solution; stop solution: 1 M hydrochloric acid.

[0067] The specific method was as follows:

[0068] (1) Coating: The antigen was diluted to 1 μg/ml with the coating solution, mixed and then added to the microtiter-plate at 100 μl per well, covered with the cover film, and placed at 4° C. overnight.

[0069] (2) Blocking: The microtiter-plate was taken out to discard the coating solution, added with the blocking solution, covered with the cover film, and incubated at 37° C. for 0.5 h.

[0070] (3) Addition of primary antibody: The anti-serum after the third immunization was first diluted 1000-fold, and then subjected to doubling dilution for 9 gradients, covered with the cover film, and incubated at 37° C. for 1 h.

[0071] (4) Addition of secondary antibody: The enzyme-labeled microtiter-plate was taken out to discard the solution inside, added with the diluted secondary antibody at a concentration of 0.033 μg/ml, covered with the cover film, and incubated at 37° C. for half an hour.

[0072] (5) Color development: The enzyme-labeled microtiter-plate was taken out to discard the solution inside, added with the chromogenic solution to develop the color at 25° C. for 13 min.

[0073] (6) Stop of reaction: The stop solution was added to stop the reaction.

[0074] (7) The value was read at 450 nm on a microtiter-plate reader immediately after the addition of the stop solution. The maximum dilution corresponding to the well having an OD value of more than 2.1 times the OD value of the set negative control was determined as the titer of the sample, and the test results are as shown in Table 1. NC is a negative control of unimmunized serum, and the initial dilution factor was 1:1,000. The anti-serum after the third immunization was tested, animal Nos. #4061, #4062, and #4063 had antiserum titers at 1:512,000, with the highest S/B values; the remaining two animals (#4064, #4065) had antiserum titers at 1:256,000, but S/B was only 2.7 or less; at the same time the sera were subjected to histochemical analysis of human neuronal cells, and all three mice had a good histochemical signal (FIG. **2**). Therefore, three animals were selected for cell fusion.

TABLE 1

Serum ELISA results after third immunization

Animal No.	#4061	#4062	#4063	#4064	#4065
Blank control	0.063	0.063	0.063	0.063	0.063
1:1,000	2.707	2.783	2.674	2.632	2.541
1:2,000	2.606	2.58	2.532	2.438	2.405
1:4,000	2.571	2.28	2.405	2.345	2.319
1:8,000	2.44	2.249	2.213	1.897	1.792
1:16,000	2.206	2.01	1.934	1.624	1.435
1:32,000	1.888	1.62	1.566	1.246	1.003
1:64,000	1.479	1.227	1.199	0.843	0.677
1:128,000	1.087	0.833	0.826	0.529	0.403
1:256,000	0.733	0.512	0.528	0.306	0.24
1:512,000	0.435	0.313	0.306	0.174	0.165
Titer	1:512,000	1:512,000	1:512,000	1:512,000	1:512,000
S/B	6.905	4.968	4.857	2.762	2.619

2.3 Confirmation of Cytologic Specificity

[0075] In order to confirm whether the sera produced by these stimulated immunizations are specific for neural tissues, immunohistochemical analyses of human neural tissues were performed on sera from five animals. The specific experimental methods were as follows.

2.3.1 Tissue Dehydration Treatment

[0076] The human nerve tissue was taken for dehydration treatment, and the dehydration treatment was performed using Leica ASP300S. The specific process was as follows:

[0077] The sections were dehydrated with 70%, 85% and 90% anhydrous ethanol for 30 minutes, respectively; then dehydrated with anhydrous ethanol for 2 times, each time for 60 minutes; then treated with a clearing agent for 30 minutes and then treated with a clearing agent 2 times for 60 minutes each; and then treated with paraffin wax 3 times for 60, 120 and 180 minutes, respectively; then subjected to the embedding operation using a Leica EG1150 embedding machine to make a wax block, which was cut into sections with a thickness of 4 μm.

2.3.2 In Situ Hybridization

[0078] The human nerve tissue sections were baked at 85° C. for 20 min; treated 3 times with a dewaxing agent for 1 minute each time; dewaxed 3 times with anhydrous alcohol for 1 minute each time; washed 3 times with water for 1 minute each time; thermal repaired with ER2 (pH-9 buffer solution) for 20 minutes, cooled for 12 minutes, then washed 3 times with water for 1 minute each time; then blocked for 30 minutes; washed 3 times with water for 1 minute each time; added with the supernatant of the cell line culture and incubated for 30 minutes, washed 3 times with water for 1 minute each time; incubated for 8 minutes with an enhancer, washed 3 times with water for 2 minutes each time, added with the secondary antibody and incubated for 8 minutes; washed 3 times with water for 2 minutes each time; developed color with DAB for 8 minutes; washed 3 times with water for 1 minute each time, stained with hematoxylin for 10 minutes; washed 3 times with water for 1 minute each time, dehydrated with alcohol, air-dried and sealed. Observations were performed using an Olympus optical microscope.

[0079] It was found by optical microscopy that the sera from all five animals reacted positively with human neural tissue, indicating that the immunized animals produced antibodies specific for nerve tissue (FIG. **2**).

2.4 Cell Fusion

[0080] According to the ELISA assay results, combined with the results of human neural tissue specific analysis, three animals #4061, #4062 and #4063 were selected for final immunization, and three days later, the spleen cells of the two animals were fused with tumor cells. The mouse myeloma cells (SP2/0) and spleen cells were electrofused in a 1:3 ratio and the fused cells were plated into 15 feeder cell plates using HAT medium, and cultured in a CO_2 incubator.

2.5 Hybridoma Cell Line Screening

[0081] After the fusion cells were cultured for 7-10 d, the whole plate was subjected to culture medium change. ELISA assay was performed after 4 h of the medium change. The specific materials and steps of ELISA were the same as those described in 2.2 for ELISA of animal serum.

[0082] 10 microtiter plates were provided for each animal, a total of 40 96-well plates, a total of 3840 wells, added fusion animal serum with 1:1000 dilution, set as a negative control; added with the blank medium, to detect the OD value of the blank control. The antibody with highest antibody titer and S/B (Signal/Blank)>=2.1 was screened and confirmed as a positive clone. A total of 135 candidate clones were obtained by screening and then used for the next round of subcloning.

2.5.1 Subcloning

[0083] The clones of 135 single-well cells from the first screening were subjected to a second assay (assay method as above). Based on the antibody titer and S/B ratio, 14 candidate clones were screened from the 135 candidate clones, and the assay results of the 14 clones are shown in Table 2. These 14 clones will be used for further screening.

2.6. Affinity Sorting

[0084] In order to screen the monoclonal antibody cell lines with relatively high affinity, 14 monoclonal antibody cell lines were subjected to affinity sorting. The results are shown in Table 3, only the three clones 5C12d2C8, 18F9G6D5 and 55F8C3B2 of the 14 monoclonal antibodies have higher confidence in the fitted curves. Among them, 5C12d2C8 had the highest affinity with a KD (M) of 4.55×10^{-9} and an Rmax (RU) of only 291.1 (Table 3), and the cell line of this clone was used as a candidate clone.

TABLE 2

Second round of subclonal screening (ELISA)												
Concentration (ng/ml)	1,000	500	250	125	62.5	31.25	15.62	7.81	3.9	1.95	Blank	Titer
Dilution factor	1:1000	1:2000	1:4000	1:8000	1:16000	1:32000	1:64000	1:128000	1:256000	1:512000	Blank	
26E7H12D8	2.825	2.658	2.609	2.497	2.195	1.986	1.572	1.224	0.835	0.54	0.063	>1:512,00
45E4D9G3	2.742	2.66	2.644	2.551	2.372	2.169	1.798	1.391	0.999	0.759	0.063	>1:512,000
12D6F3B6	2.689	2.659	2.639	2.574	2.433	2.166	1.862	1.393	0.945	0.668	0.063	>1:512,000
5C12D2C8	2.685	2.619	2.579	2.478	2.273	1.972	1.563	1.127	0.718	0.436	0.054	1:512,000
15B2B3H1	2.878	2.808	2.786	2.705	2.596	2.449	2.201	1.849	1.43	1.003	0.054	>1:512,000
20H9C1B7	2.868	2.794	2.712	2.659	2.553	2.385	2.079	1.749	3.337	0.932	0.054	>1:512,000
74D7E4A6	2.77	2.738	2.723	2.624	2.492	2.218	1.8	1.352	0.899	0.616	0.063	>1:512,000
60G12D8G5	2.854	2.727	2.607	2.5	2.272	1.997	1.552	1.067	8.724	0.464	0.054	1:512,000
64E3G5C10	2.861	2.701	2.609	2.393	2.117	1.744	1.315	0.945	0.599	0.374	0.054	1:512,000
55F8C3B2	2.745	2.668	2.593	2.466	2.218	1.956	1.544	1.063	0.74	0.481	0.063	1:512,000
68G5E5E7	2.849	2.73	2.768	2.655	2.519	2.358	2.033	1.653	1.202	0.833	0.063	>1:512,000
18F9G6D5	2.708	2.676	2.565	2.525	2.327	2.081	1.719	1.305	0.745	0.35	0.054	>1:512,000
18G5A5E2	2.64	2.625	2.52	2.435	2.287	2.023	1.642	1.218	0.794	0.491	0.054	1:512,000
60G12G2E2	2.881	2.762	2.673	2.585	2.391	2.174	1.756	1.38	0.908	0.559	0.054	>1:512,000

The titer is the highest dilution with S/B (Signal/Blank) >=2.1, the OD450 in blank is the average of two technical replicates.

TABLE 3

Antigen-antibody binding affinity sorting					
Analyte Solution	Kinetics Chi^2 (RU^2)	ka (1/Ms)	kd (1/s)	KD (M)	Rmax (RU)
5C12D2C8	1.70E+00	5.06E+04	2.30E−04	4.55E−09	291.1
60G12D8G5	2.30E+00	1.43E+06	1.41E−02	9.90E−09	36.1
68G5E5E7	7.60E−01	4.59E+06	5.45E−02	1.19E−08	19.6
20H9C1B7	3.21E+00	2.36E+05	2.91E−03	1.24E−08	26.5
60G12G2E2	4.33E−01	2.69E+06	3.38E−02	1.26E−08	16
74D7E4A6	3.31E+00	2.00E+06	2.72E−02	1.36E−08	43.1
18F9G6D5	1.03E−02	8.95E+04	1.77E−03	1.98E−08	8.9
15B1B3H1	3.02E+01	2.86E+05	7.27E−03	2.54E−08	102.8
26E7H12D8	6.97E−01	1.44E+05	5.09E−03	3.53E−08	18.2
55F8C3B2	1.39E−02	9.68E+03	4.27E−04	4.41E−08	487.4
18G5A5E2	N/A	N/A	N/A	N/A	N/A
12D6F3B6	N/A	N/A	N/A	N/A	N/A
64E5G5C10	N/A	N/A	N/A	N/A	N/A
45E4D9G3	N/A	N/A	N/A	N/A	N/A

Example 3 Antibody Immunogenicity Assay

3.1 Antigen Preparation

3.1.1: Construction of Vector and Preparation of Crude Protein

[0085] 80 amino acid polypeptides containing the ion-conducting pore module target of the DIVS3 domain of Nav1.7 were synthesized, and a His fusion protein expression vector for prokaryotic expression was constructed, and inoculated into 2000 mL of LB liquid medium (Kan-resistant), and incubated at 37° C. with shaking overnight, and then allowed to reduce the incubation temperature to 30° C. when the OD600 was about 0.6; added with IPTG inducer to a final concentration of 0.1 mM, and continued to incubate at 30° C. for 8 h; centrifuged for 3 min to collect bacteria, which were resuspended in 50 mL of pre-cooled NTA-0 buffer, and subjected to ice-bath for 30 min, and then ultrasonically fragmented, and then centrifuged for 50 min at 4° C. to collect the precipitate (inclusion bodies). The precipitate was resuspended in 50 mL of NTA-0 buffer, and added with DTT to a final concentration of 1 mM; subjected to ultrasonication to promote the dissolution of heteroproteins, centrifuged at 4° C. for 10 min to remove the supernatant, to obtain the preparation solution of crude protein.

3.1.2 Protein Denaturation and Reconstitution

[0086] The protein solution was denatured by dilution with 2 times volume of 3M guanidine hydrochloride. The protein solution was put in a dialysis bag and the volume was concentrated to 50-100 mL with PEG20000 and dialyzed at 4° C. with PBS buffer overnight for denaturation, and then the protein was concentrated.

3.1.2 Protein Purification

[0087] The Ni-NTA column was prepared according to the manufacturer's instructions, and protein purification was performed according to the instructions.

3.2 Protein Western Blotting

[0088] (1) Making separation gel according to 12% separation gel formula;

[0089] (2) Mixing the corresponding antigen with the sample buffer 4:1 and denaturing at 95° C. for 5 min;

[0090] (3) Loading 4 μl of Marker and 50 μl of antigen to the lane respectively;

[0091] (4) Running at 90V to the end of the concentrated gel, and then running to ⅔ of the end of the separated gel at 120V;

[0092] (5) Taking out the gel to remove the concentrated gel, and putting the cut PVDF membrane in methanol for 2 min, and then equilibrating the gel, PVDF membrane, 6 cut filter paper and a supporting pad in a transfer buffer for 10 min;

[0093] (6) Assembling in the order of negative electrode→blackboard→supporting pad→3 filter papers→glue→membrane→3 filter papers→supporting pad→whiteboard→positive electrode, and driving out air bubbles with a glass rod in each layer;

[0094] (7) Filling the membrane transfer tank with membrane transfer buffer, transfering the membrane at 100V for 30 min, and put the membrane transfer tank into ice water;

[0095] (8) Taking out the membrane and staining it with Reichhorn red dye for 5 min, then rinsing it slightly with water to observe whether the membrane transfer is successful, and washing the Reichhorn red with washing solution after the appearance of bands;

[0096] (9) Moistening the membrane with blocking solution, then blocking it on shaking bed for 1 h, then washing with washing solution for 3 times, 5 min each;

[0097] (10) Cutting the membrane into 5 strips, taking 5 different hybridoma supernatants, diluting them with PBS at a ratio of 1:10 as primary antibody, and incubating at 4° C. overnight;

[0098] (11) On the next day, washing with washing solution for 3 times, 5 min each, adding HRP-sheep anti-mouse IgG diluted (1:5000) with washing solution and incubating for 1h at room temperature;

[0099] (12) Washing with washing solution for three times, 10 min each, dropring the prepared DAB working solution onto the membrane, and observing the results after color development in the dark for 8 min.

4. Western Blotting Results

[0100] The monoclonal antibody 5C12D2C8 to Nav1.7 was hybridized with the covered target peptide antigen by Western Blotting, and the results are shown in FIG. **3**. The results showed that 5C12D2C8 can recognize the target polypeptide antigen with antigen-antibody immunoreaction, proving that 5C12D2C8 is able to recognize the target and has good specificity.

Example 4 Sequencing of Antibodies

[0101] To determine the monoclonal antibody sequence, monoclonal 5C12D2C8 was sequenced. Total RNA was isolated from hybridoma cells according to the technical manual for TRIzol reagent. Total RNA was then reverse transcribed into cDNA using isoform-specific antisense primers or universal primers, following the PrimeScript TM First Strand cDNA Synthesis Kit Technical Manual. Antibody fragments for VH and VL were amplified according to GenScript's Rapid Amplification of cDNA Ends (RACE) standard operating procedure (SOP) method. The amplified antibody fragments were cloned into standard cloning vectors, respectively. Colony PCR was performed to screen the clones with insert fragments of the correct size. At least 5 colonies with correctly sized insert fragments were sequenced. The sequences of the different clones were compared to determine the common sequence of these clones.

[0102] Finally, the DNA sequence of VH was determined as shown in SEQ ID NO: 10; the DNA sequence of VL was determined as shown in SEQ ID NO: 11.

[0103] The DNA sequence of the heavy chain variable region (VH) of monoclonal antibody 5C12D2C8 is shown in SEQ ID NO: 10; the DNA sequence of the light chain variable region (VL) is shown in SEQ ID NO: 11. VH and VL contain three complementary determining regions (CDRs) respectively, each CDR positions as the following: the dashed line underlined is the guide sequence (signal peptide), the solid line underlined is the CDR sequence, and the boldface is the framework region (FR).

Heavy chain:

Guide sequence-**FR1**-CDR1-**FR2**-CDR2-**FR3**-CDR3-**FR4**-constant region-Stop codon

Heavy chain variable region (VH):

(SEQ ID NO: 10)

ATGAACTTCGGGCTCAGCTTGATTTTCCTTGCCCTCATTTTAAAAGGTGTCC

AGTGT**GAGGTGCAGCTGGTGGAGTCTGGGGGAGACTTAGTGAAGCCT**

GGAGGGTCCCTGAAACTCTCCTGTGCAGCCTCTGGATTCACTTTCAG

TAGCGGCGGCATGTCT**TGGGTTCGCCAGACTCCAGACAAGAGGCTGGA**

GTGGGTCGCAACCATTAGTAATGGTGGTGGTGAGACCTACTATGAGGACA

GTGTGAAGGGG**CGATTCACCATCTCCAGAGACAATGCCAAGAACACCC**

TGTACCTACAAATGAACAGTCTGAAGTCTGAGGACACAGCCAT

GTATTACTGTGCTAGGGAGGAGTACGAGGGTACGTGGTACTGAGATGTCT

GGGGCGCAGGGACCACGGTCACCGTCTCCTCA

Heavy chain constant region:
GCCAAAACGACACCCCCATCTGTCTATCCACTGGCCCCTGGATCTGCTGCC
C
AAACTAACTCCATGGTGACCCTGGGATGCCTGGTCAAGGGCTATTTCCCTG
AGCCAGTGACAGTGACCTGGAACTCTGGATCCCTGTCCAGCCGTGTGCAC
ACCTTCCCACCTCTCCTCCAGTCTGACCTCTAC
ACTCTGAGCAGCTCAGTGACTGTCCCCTCCAGCACCTGGCCCAGCGAGAC
CGTCACCTGCAACGTTG
CCCACCCGGCCAGCAGCACCAAGGTGGACAAGAAAATTGTGCOCAGGGA
TIGTGGTTGTAAGCCTTG
CATATGTACAGTCCCAGAAGTATCATCTGTCTTCATCTTCCCCCCAAAGCCC
AAGGATGTGCTCACC
ATTACTCTGACTCCTAAGGTCACGTGTGTTGTGGTAGACATCAGCAAGGAT
GATCCCGAGGTCCAGT
TCAGCTGGTTTGTAGATGATGTGGAGGTGCACACAGCTCAGACGCAACCC
CGGGAGGAGCAGTTCAA
CAGCACTTTCCGCTCAGTCAGTGAACTTCCCATCATGCACCAGGACTGGCT
CAATGGCAAGGAGTTC
AAATGCAGGGTCAACAGTGCAGCTTTCCCTGCCCCCATCGAGAAAACCAT
CTCCAAAACCAAAGGCA
GACCGAAGGCTCCACAGGTGTACACCATTCCACCTCCCAAGGAGCAGATG
GCCAAGGATAAAGTCAG
TCTGACCTGCATGATAACAGACTTCTTCCCTGAAGACATTACTGTGGAGTG
GCAGTGGAATGGGCAG
CCAGOGGAGAACTACAAGAACACTCAGCCCATCATGGACACAGATGGCTC
TTACTTCGTCTACAGCA
AGCTCAATGTGCAGAAGAGCAACTGGGAGGCAGGAAATACTTTCACCTGC
TCTGTGTTACATGAGGG
CCTGCACAACCACCATACTGAGAAGAGCCTCTCCCACTCTCCTOGTAAATG
A

Light chain:

Signal peptide-**FR1**-CDR1-**FR2**-CDR2-**FR3**-CDR3-**FR4**-Constant region-Stop codon

-continued

Light chain variable region (VL):

(SEQ ID NO: 11)

ATGAAGTTGCCTGTTAGGCTGTTGGTGCTGATGTTCTGGATTCCTGCTTCCA

GCAGT**GATGTTTTGATGACCCAAACTCCACTCTCCCTGCCTGTCAGTC**

TTGGAGATCAAGCCTCCATCTCTTGCACATCTAGTCAGAGCGGAGTACA

TAGTAATGGAAACACCGAGTTAGAA**TGGTACCTGCAGAAACCAGGCCA**

GTCT

CCAAAGCTCCTGATCTACAAAGTTTCCAACCGAACATCT**GGGGTCCCAG**

ACAGGTTCAGTGGCAGTG

GATCAGGGACAGATTTCACACTCAAGATCAGCAGAGTGGAGGCTGAG

GATCTGGGAGTTTATTACTGCGGCCAAGGTTCACATGTTGGCCGGACG**T**

TCGGTGGAGGCACCAAGCTGGAAATCAAA

Light chain constant region:
CGGGCTGATGCTGCACCAACTGTATCCATCTTCCCACCATCCAGTGAGCAG
TTAACATCTGGAGGTGCCTCAGTCGTGTGCTTCTTGAACAACTTCTACCCC
AAAGACATCAATGTCAAGTGGAAGATTGATGGCAGIGAACG
ACAAAATGGCGTCCTGAACAGTTGGACTGATCAGGACAGCAAAGACAGC
ACCTACAGCATGAGCAOC
ACCCTCACGTTGACCAAGGACGAGTATGAACGACATAACAGCTATACCTGT
GAGGCCACTCACAAGA
CATCAACTTCACCCATTOTCAAGAGCTTCAACAGGAATGAGTGTTAG

[0104] The amino acid sequence was deduced from the DNA sequence. The amino acid sequence of VH is shown in SEQ ID NO: 7 and the amino acid sequence of VL is shown in SEQ ID NO: 8.

Heavy chain:

Signal peptide-FR1-CDR1-FR2-CDR2-FR3-CDR3-FR4-Constant region-Stop

codon

Heavy chain variable region (VH):

(SEQ ID NO:7)

MNFGLSLIFLALILKGVQC**EVQLVESGGDLVKPGGSLKLSCAASGFTFS**SGG

MS**WVRQTPDKRLEWVA**TISNGGGETYYEDSVKG**RFTISRDNAKNTLYLQ**

MNSLKSEDTAMYYCAREEYEGTWYEDV**WGAGTTVTVSS**

Constant region

AKTTPPSVYPLAPGSAAQTNSMVTLGCLVKGYFPEPVTVTWNSGSLSSGVHT

FPAVLQSDLYTLSSSVTVPSSTWPSETVTCNVAHPASSTKVDKKIVPRDCGCKP

CICTVPEVSSVFIFPPKPKDVLT

ITLTPKVTCVVVDISKDDPEVQFSWFVDDVEVHTAQTQPREEQFENSTERSVSE

LPIMHQDWLNGKEF

KCRVNSAAFPAPIEKTISKTKGRPKAPQVYTIPPPKEQMAKDKVSLTCMITDFF

PEDITVEWQWNGQ

-continued

PAENYKNTQPIMDTDGSYFVYSKLNVOKSNWEAGNTFTCSVLHEGLHNHHT

EKSLSHSPGK-
Light chain:
Signal peptide-**FR1**-CDR1-**FR2**-CDR2-**FR3**-CDR3-**FR4**-Constant region-Stop codon

Light chain variable region (VL)

(SEQ ID NO:8)

MKLPVRLLVLMFWIPASSS**DVLMTQTPLSLPVSLGDQASISC**TSSQSGVH
SNGNTELE**WYLQKPGQSPKLLIY**KVSNRTS**GVPDRFSGSGSGTDFTLKL**
SRVEAEDLGVYYCGQGSHVGRT**FGGGTKL**
KEI
Constant region
RADAAPTVSIFPPSSEQLTSGGASVVCFLNNFYPKDINVKWKIDGSERQNG
VLNSWTDQDSK
DSTYSMSSTLTLTKDEYERHNSYTCEATHKTSTSPIVKSFNRNEC-

[0105] It was deduced that the RNA sequence encoding VH is shown in SEQ ID NO: 12 and the RNA sequence encoding VL is shown in SEQ ID NO: 13.

Example 5 Determination of Affinity of 5C12D2C8 Antibody to Antigen

[0106] The antigen-antibody dissociation rate equilibrium constant (KD) reflects the affinity between the antibody and the antigen, and the lower the KD value, the higher the affinity. Surface plasmon resonance (SPR) was used to determine the antigen-antibody dissociation rate equilibrium constant of a monoclonal antibody, to evaluate the affinity of the monoclonal antibody to the antigen.

[0107] The binding curves of monoclonal antibody 5C12D2C8 were determined by SPR at different concentrations of 12.5, 25, 50, 100, 200 and 400 nM. The results are shown in FIG. **5**. The dissociation constant Kd (l/s) of monoclonal antibody 5C12D2C8 to antigen was 2.47×10^{-4} and the equilibrium dissociation constant KD (M) was 8.78×10^{-9} (FIG. **4**).

Example 6 Specificity of 5C12D2C8 Antibody in Human Peripheral Nerve Cells

[0108] Since nerve cells are highly differentiated, in order to test the specificity of 5C12D2C8 in human peripheral nerve cells, iPS-induced human peripheral nerve cells were used for validation. Before performing the test, the type of iPS-induced neuronal cells was confirmed, molecular markers specific for peripheral nerve tissues were used to identify whether the induced peripheral nerve well developed. Three molecular markers specifically expressed in neuronal cells, PSD95, were used to localize human nerve tissues. PSD95 is encoded by the DLG4 gene, a member of the membrane-associated guanylate kinase (MAGUK) family, which interacts with PSD93 at postsynaptic sites and is specifically expressed in nerve tissues. Therefore, the PSD95 antibody can be used as a molecular marker (control) to study the specificity and distribution of antibody expression in human peripheral nerve cells.

6.1 iPS Induction of Human Peripheral Neurons

[0109] 6.1.1: T25 culture flasks were coated with 2 μg/cm^2 of laminin (SIGMA); human-derived nerve stem cells (iRegene Therapeutics) were inoculated into 2 coated T25 culture flasks at 2.5×10^6 cells per flask in serum-free, animal origin-free FP neural stem cell medium (iRegene Therapeutics) where 10 UM of inhibitor Y-27632 was added. After inoculation, the T25 culture flasks were incubated overnight at 37° C., 5% CO_2.

[0110] 6.1.2: The following day, the FP neural stem cell medium containing Y-27632 was removed and replaced with serum-free, animal origin-free peripheral neuronal directed differentiation medium (iRegene Therapeutics), and this medium was then used until the end of the experiment. The medium was then changed every other day until day 14; cell morphology changes were recorded in between.

[0111] 6.1.3: Double-coated 15 mm optical slides (Deckglaser) were used to prepare cell climbing slides. 15 mm optical slides were placed in 24-well plates (1 slide/well). The slides were submerged with 50 μg/mL poly-L-lysine solution and incubated at 37° C., 5% CO_2 for no less than 2 hours. The poly-L-lysine was removed and washed three times with sterilized water. The slides were submerged using 5 μg/mL of Laminin solution and incubated at 37° C., 5% CO_2 for not less than 3 hours, after which they were washed well with PBS.

[0112] 6.1.4: Cells in T25 culture flasks were digested using Accutase Cell Digest (Invitrogen) and cells were inoculated onto the coated 15 mm optical slides (Deckglaser) at 1×10^5 cell/well. The medium was serum-free, animal origin-free neuronal directed differentiation medium (iRegene Therapeutics), and the medium was changed every other day until day 21.

6.2 Fluorescence Immunoassay

[0113] 6.2.1: The medium for cell differentiation was removed from the 24-well microtiter plates and the culture wells were washed using DPBS.

[0114] 6.2.2: Cells were fixed using 4% paraformaldehyde for 40 minutes at room temperature and washed twice with DPBS buffer; then treated with 0.1% Triton X-100 for 5 minutes and then washed twice with DPBS buffer. The cells were then incubated with DPBS buffer containing 10% horse serum and 0.1% Triton X-100 at 4° C. overnight; finally, the antibody diluted with DPBS buffer was added and incubated for 2 hours at 37° C. and washed three times with DPBS buffer; Afterwards, the cells were incubated with fluorescent secondary antibody (1:1000) corresponding to the primary antibody, and the secondary antibody was removed after 45 min and washed three times with DPBS buffer; nuclear staining was performed using 300 nM DAPI for 2 minutes at room temperature; optical cell climbing slides were prepared by washing the seals three times with sterilized water; and read with an OLYMPUS FV3000 laser confocal microscope or a NIKON N-SIM high-resolution microscope.

6.3 Localization Results of Human Peripheral Nerve Cells

[0115] The cell type analysis of iPS-induced peripheral nerve cells was carried out using the peripheral nerve cell-specific molecular marker PSD95. The results showed that iPS-induced peripheral nerve cells can bind to the specific molecular marker, which proves that iPS-induced nerve cells belong to peripheral nerve cells, and can be used for cell-specific analysis of antibodies. The distribution of monoclonal antibody 5C12D2C8 in iPS-induced human peripheral nerve cells is shown in FIG. **5**. Monoclonal antibody 5C12D2C8 showed stronger fluorescence signal on axons of peripheral nerve cells, compared with the control PSD95. The results showed that Nav1.7 channel monoclonal antibody 5C12D2C8 co-localized with molecular markers specific to axonal molecules, indicating that 5C12D2C8 has good peripheral nerve cell specificity and is aligned in dots on axons of peripheral nerve cells.

Example 7 Analgesic Effect of 5C12D2C8 Antibody in Wild-type Mice

[0116] When injury receptors are stimulated, action potentials are generated and transmitted through peripheral nerves to the spinal cord and then to the brain to feel pain, and the Nav1.7 voltage-gated sodium channel plays a crucial role in the generation and transmission of action potentials. We hypothesize that monoclonal antibody drugs specifically bind to the ion-conducting pore of the voltage-gated sodium channel Nav1.7, which may block electrical signaling and thus achieve analgesia.

[0117] In order to validate and detect the analgesic effect of the monoclonal antibody 5C12D2C8, the formalin inflammatory pain model was used in this study. Since the pain caused by formalin experiments is more oriented to tension pain and elicits a state very similar to clinical pain, which is widely used to determine the pain perception of animals. After injection of formalin, the response behavior of animals to injurious stimuli are presented in 2 periods, the first period arises immediately after injection and lasts for about 10 min, and the second period appears about 15 min after injection and lasts for about 45 min, and thus the 2 periods are named as phase I and phase II, respectively. It is generally believed that phase I pain is caused by the direct stimulation of injury receptors by formalin, which leads to the excitation of C fibers and thus produces pain; while phase II pain is caused by inflammation-induced release of neurotransmitters such as prostaglandins, histamine, and 5-hydroxytryptamine, which excites the nociceptive nerve to produce pain. It is generally believed that phase II is more responsive to the effects of the drugs.

7.1 Experimental Methods

7.1.1: Experimental Apparatus

[0118] Mouse tail injection fixator (YLS-Q9G) was purchased from Shanghai Bio-will Co., Ltd. and 50 μl micro syringe and 1 ml human insulin syringe were purchased from Wuhan Qinzhijie Biological Co., Ltd.

7.1.2: Experimental Animal 25-35 g SPF grade KM mice were purchased from Hubei Animal Experiment and Research Center; KM mice were purchased to acclimatize in the laboratory for at least 2 days, with an ambient temperature of 23±1° C., and were given adequate water as well as food, 5-8 mice per cage. The mice were acclimatized in transparent cages for 30 minutes prior to the experiment. No food or water was allowed during the experiment. A timer was used to record the results, and at the end of the experiment the animals were anesthetized and euthanized by placing them in ether vials.

7.1.3: Drug Treatment Method

[0119] Each mouse was taken out from the transparent cage and placed on the tabletop and put into a fixator by pulling its tail. Then the head was fixed with a stopper so that its tail extended out of the bottom of the fixator. The fixator was placed on the tail injection auxiliary table, and the tail of the mouse was obsereved under the auxiliary lamp. After the lateral blood vessels were observed, 200 μl of monoclonal antibody drug was injected into the lateral blood vessels by tail vein injection, after the needle was pulled out, the wound was pressed for 20 s to stop bleeding, and then the mouse was placed into the transparent cage. After 30 min, 20 μl of freshly prepared 5% formalin solution was injected into the plantar surface of the mouse using a microsyringe, and then the mouse was immediately placed in the transparent cage for observation. A timer was used to record the time(s) for the mouse to lick the injected paw over a 45-min period, and the data were recorded at 5-min intervals, by recording the time of paw licking and paw retraction during each 5-min period, for a total of 45 min. Phase I acute pain (0-10 min) and phase II persistent pain (10-45 min) were statistically analyzed, respectively, using murine IgG control antibody with equal volume and concentration as the control group.

7.1.4: Data Processing

[0120] The data were subjected to two-tailed T-test to determine if there was a significant difference. In the same dose experiment, the data of the experimental group was subjected to two-tailed T-test with both PBS control and IgG control; in different dose experiments, the data of the drug experimental group were compared with the PBS control group as well as equal dose IgG control group and also between different doses of the same drug. Significant differences were indicated by *, where * indicated $p<0.05$ and ** indicated $p<0.01$.

7.2 Experimental Results

[0121] The analgesic effect of monoclonal antibody 5C12D2C8 at a dose of 25 mg/kg is shown in FIG. **6**. IgG negative antibody control (IgG) showed a significant increase in paw licking time in phase II over phase I at the same dose. For the mice injected with monoclonal antibody 5C12D2C8, the paw licking time of in phase II was significantly different (P<0.01) from that in phase I and in control group after administration. The analgesic effect of monoclonal antibody 5C12D2C8 in phase II was reduced by 32% compared to the negative antibody control (IgG) (FIG. **6**).

Sequence Listing:

HCDR 1:

(SEQ ID NO: 1)

SGGMS

HCDR2:

(SEQ ID NO: 2)

TISNGGGETYYEDSVKG

HCDR 3:

(SEQ ID NO: 3)

EEYEGTWYEDV

LCDR 1:

(SEQ ID NO: 4)

TSSQSGVHSNGNTELE

LCDR2:

(SEQ ID NO: 5)

KVSNRTS

LCDR 3:

(SEQ ID NO: 6)

GQGSHVGRT

VH:

(SEQ ID NO: 7)

MNFGLSLIFLALILKGVQCEVQLVESGGDLVKPGGSLKLSCAASG

FTFSSGGMSWVRQTPDKRLEWATISNGGGETYYEDSVKGRFTISR

DNAKNTLYLQMNSLKSEDTAMYYCAREEYEGTWYEDVWGAGTVTV

SS

Heavy chain:

(SEQ ID NO: 14)

MNFGLSLIFLALILKGVQCEVQLVESGGDLVKPGGSLKLSCAASG

FTFSSGGMSWVRQTPDKRLEWATISNGGGETYYEDSVKGRFTISR

DNAKNTLYLQMNSLKSEDTAMYYCAREEYEGTWYEDVWGAGTAKT

TPPSVYPLAPGSAAQTNSMVTLGCLVKGYFPEPVTVTWNSGSLSS

GVHTFPAVLQSDLYTLSSSVTVPSSTWPSETVTCNVAHPASSTKV

DKKIVPRDCGCKPCICTVPEVSSVFIFPPKPKDVLTITLTPKVTC

VVVDISKDDPEVQFSWFVDDVEVHTAQTQPREEQFNSTFRSVSEL

PIMHQDWLNGKEFKCRVNSAAFPAPIEKTISKTKGRPKAPQVYTI

PPPKEQMAKDKVSLTCMITDFFPEDITVEWQWNGQPAENYKNTQP

IMDTDGSYFVYSKLNVQKSNWEAGNTFTCSVLHEGLHNHHTEKSL

SHSPGK-

VL:

(SEQ ID NO: 8)

MKLPVRLLVLMFWIPASSSDVLMTQTPLSLPVSLGDQASISCTSS

QSGVHSNGNTELEWYLQKPGQPKLLIYKVSNRTSGVPDRFSGSGS

GTDFTLKISRVEAEDLGVYYCGQGSHVGRTFGGGTKLEI

-continued

Light chain:

(SEQ ID NO: 15)

MKLPVRLLVLMFWIPASSSDVLMTQTPLSLPVSLGDQASISCTSS

QSGVHSNGNTELEWYLQKPGQPKLLIYKVSNRTSGVPDRFSGSGS

GTDFTLKISRVEAEDLGVYYCGQGSHVGRTFGGGTKLEIKRAAAP

TVSIFPPSSEQLTSGGASVVCFLNNFYPKDINVKWKIDGSERQNG

VLNSWTDQDSKDSTYSMSSTLTLTKDEYERHNSYTCEATHKTSTS

PIVKSFNRNEC-

Antigen peptide:

(SEQ ID NO: 9)

DSVNVDKQPKYEYS

VH nucleotide sequence:

(SEQ ID NO: 10)

atgaacttcgggctcagcttgattttccttgccctcattttaaaa

ggtgtccagtgtgaggtgcagctggtggagtctgggggagactta

gtgaagcctggagggtccctgaaactctcctgtgcagcctctgga

ttcactttcagtagcggcggcatgtcttgggttcgccagactcca

gacaagaggctggagtgggtcgcaaccattagtaatggtggtggt

gagacctactatgaggacagtgtgaaggggcgattcaccatctcc

agagacaatgccaagaacaccctgtacctacaaatgaacagtctg

aagtctgaggacacagccatgtattactgtgctagggaggagtac

gagggtacgtggtactgagatgtctggggcgcagggaccacggtc

accgtctcctca

Heavy chain nucleotide sequence:

(SEQ ID NO: 16)

atgaacttcgggctcagcttgattttccttgccctcattttaaaa

ggtgtccagtgtgaggtgcagctggtggagtctgggggagactta

gtgaagcctggagggtccctgaaactctcctgtgcagcctctgga

ttcactttcagtagcggcggcatgtcttgggttcgccagactcca

gacaagaggctggagtgggtcgcaaccattagtaatggtggtggt

gagacctactatgaggacagtgtgaaggggcgattcaccatctcc

agagacaatgccaagaacaccctgtacctacaaatgaacagtctg

aagtctgaggacacagccatgtattactgtgctagggaggagtac

gagggtacgtggtactgagatgtctggggcgcagggaccacggtc

accgtctcctcagccaaaacgacacccccatctgtctatccactg

gcccctggatctgctgcccaaactaactccatggtgaccctggga

tgcctggtcaagggctatttccctgagccagtgacagtgacctgg

aactctggatccctgtccagcggtgtgcacaccttcccagctgtc

ctgcagtctgacctctacactctgagcagctcagtgactgtcccc

tccagcacctggcccagcgagaccgtcacctgcaacgttgcccac

ccggccagcagcaccaaggtggacaagaaaattgtgcccagggat

tgtggttgtaagccttgcatatgtacagtcccagaagtatcatct

gtcttcatcttccccccaaagcccaaggatgtgctcaccattact

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ctgactcctaaggtcacgtgtgttgtggtagacatcagcaaggat

gatcccgaggtccagttcagctggtttgtagatgatgtggaggtg

cacacagctcagacgcaaccccgggaggagcagttcaacagcact

ttccgctcagtcagtgaacttcccatcatgcaccaggactggctc

aatggcaaggagttcaaatgcagggtcaacagtgcagctttccct

gcccccatcgagaaaaccatctccaaaaccaaaggcagaccgaag

gctccacaggtgtacaccattccacctcccaaggagcagatggcc

aaggataaagtcagtctgacctgcatgataacagacttcttccct

gaagacattactgtggagtggcagtggaatgggcagccagcggag

aactacaagaacactcagcccatcatggacacagatggctcttac

ttcgtctacagcaagctcaatgtgcagaagagcaactgggaggca

ggaaatactttcacctgctctgtgttacatgagggcctgcacaac

caccatactgagaagagcctctcccactctcctggtaaatga

VL nucleotide sequence:

(SEQ ID NO: 11)

atgaagttgcctgttaggctgttggtgctgatgttctggattcct

gcttccagcagtgatgttttgatgacccaaactccactctccctg

cctgtcagtcttggagatcaagcctccatctcttgcacatctagt

cagagcggagtacatagtaatggaaacaccgagttagaatggtac

ctgcagaaaccaggccagtctccaaagctcctgatctacaaagtt

tccaaccgaacatctggggtcccagacaggttcagtggcagtgga

tcagggacagatttcacactcaagatcagcagagtggaggctgag

gatctgggagtttattactgcggccaaggttcacatgttggccgg

acgttcggtggaggcaccaagctggaaatcaaa

Light chain nucleotide sequence:

(SEQ ID NO: 17)

atgaagttgcctgttaggctgttggtgctgatgttctggattcct

gcttccagcagtgatgttttgatgacccaaactccactctccctg

cctgtcagtcttggagatcaagcctccatctcttgcacatctagt

cagagcggagtacatagtaatggaaacaccgagttagaatggtac

ctgcagaaaccaggccagtctccaaagctcctgatctacaaagtt

tccaaccgaacatctggggtcccagacaggttcagtggcagtgga

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tcagggacagatttcacactcaagatcagcagagtggaggctgag

gatctgggagtttattactgcggccaaggttcacatgttggccgg

acgttcggtggaggcaccaagctggaaatcaaacgggctgatgct

gcaccaactgtatccatcttcccaccatccagtgagcagttaaca

tctggaggtgcctcagtcgtgtgcttcttgaacaacttctacccc

aaagacatcaatgtcaagtggaagattgatggcagtgaacgacaa

aatggcgtcctgaacagttggactgatcaggacagcaaagacagc

acctacagcatgagcagcaccctcacgttgaccaaggacgagtat

gaacgacataacagctatacctgtgaggccactcacaagacatca

acttcacccattgtcaagagcttcaacaggaatgagtgttag

RNA sequences:

(SEQ ID NO: 12)

AUGAACUUCGGGCUCAGCUUGAUUUUCCUUGCCCUCAUUUUAAAA

GGUGUCCAGUGUGAGGUGCAGCUGGUGGAGUCUGGGGGAGACUUA

GUGAAGCCUGGAGGGUCCCUGAAACUCUCCUGUGCAGCCUCUGGA

UUCACUUUCAGUAGCGGCGGCAUGUCUUGGGUUCGCCAGACUCCA

GACAAGAGGCUGGAGUGGGUCGCAACCAUUAGUAAUGGUGGUGGU

GAGACCUACUAUGAGGACAGUGUGAAGGGGCGAUUCACCAUCUCC

AGAGACAAUGCCAAGAACACCCUGUACCUACAAAUGAACAGUCUG

AAGUCUGAGGACACAGCCAUGUAUUACUGUGCUAGGGAGGAGUAC

GAGGGUACGUGGUACUGAGAUGUCUGGGGCGCAGGGACCACGGUC

ACCGUCUCCUCA

(SEQ ID NO: 13)

AUGAAGUUGCCUGUUAGGCUGUUGGUGCUGAUGUUCUGGAUUCCU

GCUUCCAGCAGUGAUGUUUUGAUGACCCAAACUCCACUCUCCCUG

CCUGUCAGUCUUGGAGAUCAAGCCUCCAUCUCUUGCACAUCUAGU

CAGAGCGGAGUACAUAGUAAUGGAAACACCGAGUUAGAAUGGUAC

CUGCAGAAACCAGGCCAGUCUCCAAAGCUCCUGAUCUACAAAGUU

UCCAACCGAACAUCUGGGGUCCCAGACAGGUUCAGUGGCAGUGGA

UCAGGGACAGAUUUCACACUCAAGAUCAGCAGAGUGGAGGCUGAG

GAUCUGGGAGUUUAUUACUGCGGCCAAGGUUCACAUGUUGGCCGG

ACGUUCGGUGGAGGCACCAAGCUGGAAAUCAAA

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 17

<210> SEQ ID NO 1
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 1

Ser Gly Gly Met Ser

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```
1               5


<210> SEQ ID NO 2
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 2

Thr Ile Ser Asn Gly Gly Gly Glu Thr Tyr Tyr Glu Asp Ser Val Lys
1               5                   10                  15

Gly


<210> SEQ ID NO 3
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 3

Glu Glu Tyr Glu Gly Thr Trp Tyr Glu Asp Val
1               5                   10


<210> SEQ ID NO 4
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 4

Thr Ser Ser Gln Ser Gly Val His Ser Asn Gly Asn Thr Glu Leu Glu
1               5                   10                  15


<210> SEQ ID NO 5
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 5

Lys Val Ser Asn Arg Thr Ser
1               5


<210> SEQ ID NO 6
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 6

Gly Gln Gly Ser His Val Gly Arg Thr
1               5


<210> SEQ ID NO 7
<211> LENGTH: 137
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: heavy chain variable region

<400> SEQUENCE: 7

Met Asn Phe Gly Leu Ser Leu Ile Phe Leu Ala Leu Ile Leu Lys Gly
1               5                   10                  15

Val Gln Cys Glu Val Gln Leu Val Glu Ser Gly Gly Asp Leu Val Lys
            20                  25                  30

Pro Gly Gly Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe
```

-continued

```
        35                  40                  45

Ser Ser Gly Gly Met Ser Trp Val Arg Gln Thr Pro Asp Lys Arg Leu
    50                  55                  60

Glu Trp Ala Thr Ile Ser Asn Gly Gly Gly Glu Thr Tyr Tyr Glu Asp
65                  70                  75                  80

Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr
                85                  90                  95

Leu Tyr Leu Gln Met Asn Ser Leu Lys Ser Glu Asp Thr Ala Met Tyr
            100                 105                 110

Tyr Cys Ala Arg Glu Glu Tyr Glu Gly Thr Trp Tyr Glu Asp Val Trp
        115                 120                 125

Gly Ala Gly Thr Val Thr Val Ser Ser
    130                 135


<210> SEQ ID NO 8
<211> LENGTH: 129
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: light chain variable region

<400> SEQUENCE: 8

Met Lys Leu Pro Val Arg Leu Leu Val Leu Met Phe Trp Ile Pro Ala
1               5                   10                  15

Ser Ser Ser Asp Val Leu Met Thr Gln Thr Pro Leu Ser Leu Pro Val
            20                  25                  30

Ser Leu Gly Asp Gln Ala Ser Ile Ser Cys Thr Ser Ser Gln Ser Gly
        35                  40                  45

Val His Ser Asn Gly Asn Thr Glu Leu Glu Trp Tyr Leu Gln Lys Pro
    50                  55                  60

Gly Gln Pro Lys Leu Leu Ile Tyr Lys Val Ser Asn Arg Thr Ser Gly
65                  70                  75                  80

Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
                85                  90                  95

Lys Ile Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Tyr Cys Gly
            100                 105                 110

Gln Gly Ser His Val Gly Arg Thr Phe Gly Gly Gly Thr Lys Leu Glu
        115                 120                 125

Ile


<210> SEQ ID NO 9
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: polypeptide

<400> SEQUENCE: 9

Asp Ser Val Asn Val Asp Lys Gln Pro Lys Tyr Glu Tyr Ser
1               5                   10


<210> SEQ ID NO 10
<211> LENGTH: 417
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: VH
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-continued

<400> SEQUENCE: 10

```
atgaacttcg ggctcagctt gattttcctt gccctcattt taaaaggtgt ccagtgtgag     60
gtgcagctgg tggagtctgg gggagactta gtgaagcctg gagggtccct gaaactctcc    120
tgtgcagcct ctggattcac tttcagtagc ggcggcatgt cttgggttcg ccagactcca    180
gacaagaggc tggagtgggt cgcaaccatt agtaatggtg gtggtgagac ctactatgag    240
gacagtgtga aggggcgatt caccatctcc agagacaatg ccaagaacac cctgtaccta    300
caaatgaaca gtctgaagtc tgaggacaca gccatgtatt actgtgctag ggaggagtac    360
gagggtacgt ggtactgaga tgtctggggc gcagggacca cggtcaccgt ctcctca       417
```

<210> SEQ ID NO 11
<211> LENGTH: 393
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: VL

<400> SEQUENCE: 11

```
atgaagttgc ctgttaggct gttggtgctg atgttctgga ttcctgcttc cagcagtgat     60
gttttgatga cccaaactcc actctccctg cctgtcagtc ttggagatca agcctccatc    120
tcttgcacat ctagtcagag cggagtacat agtaatggaa acaccgagtt agaatggtac    180
ctgcagaaac caggccagtc tccaaagctc ctgatctaca aagtttccaa ccgaacatct    240
ggggtcccag acaggttcag tggcagtgga tcagggacag atttcacact caagatcagc    300
agagtggagg ctgaggatct gggagtttat tactgcggcc aaggttcaca tgttggccgg    360
acgttcggtg gaggcaccaa gctggaaatc aaa                                 393
```

<210> SEQ ID NO 12
<211> LENGTH: 417
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: RNA sequence encoding VH

<400> SEQUENCE: 12

```
augaacuucg ggcucagcuu gauuuuccuu gcccucauuu uaaaaggugu ccagugugag     60
gugcagcugg uggagucugg gggagacuua gugaagccug gagggucccu gaaacucucc    120
ugugcagccu cuggauucac uuucaguagc ggcggcaugu cuuggguucg ccagacucca    180
gacaagaggc uggagugggu cgcaaccauu aguaauggug gugguggagac cuacuaugag    240
gacagaguga aggggcgauu caccaucucc agagacaaug ccaagaacac ccuguaccua    300
caaaugaaca gucugaaguc ugaggacaca gccauguauu acugugcuag ggaggaguac    360
gaggguacgu gguacugaga ugucuggggc gcagggacca cggucaccgu cuccuca       417
```

<210> SEQ ID NO 13
<211> LENGTH: 393
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: RNA sequence encoding VL

<400> SEQUENCE: 13

```
augaaguugc cuguuaggcu guuggugcug auguucugga uuccugcuuc cagcagugau     60
```

-continued

```
guuuugauga cccaaacucc acucucccug ccugucaguc uuggagauca agccuccauc    120

ucuugcacau cuagucagag cggaguacau aguaauggaa acaccgaguu agaaugguac    180

cugcagaaac caggccaguc uccaaagcuc cugaucuaca aaguuuccaa ccgaacaucu    240

ggggucccag acagguucag uggcagugga ucagggacag auuucacacu caagaucagc    300

agaguggagg cugaggaucu gggaguuuau uacugcggcc aagguucaca uguuggccgg    360

acguucggug gaggcaccaa gcuggaaauc aaa                                 393


&lt;210&gt; SEQ ID NO 14
&lt;211&gt; LENGTH: 456
&lt;212&gt; TYPE: PRT
&lt;213&gt; ORGANISM: Artificial Sequence
&lt;220&gt; FEATURE:
&lt;223&gt; OTHER INFORMATION: heavy chain

&lt;400&gt; SEQUENCE: 14

Met Asn Phe Gly Leu Ser Leu Ile Phe Leu Ala Leu Ile Leu Lys Gly
1               5                   10                  15

Val Gln Cys Glu Val Gln Leu Val Glu Ser Gly Gly Asp Leu Val Lys
            20                  25                  30

Pro Gly Gly Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe
        35                  40                  45

Ser Ser Gly Gly Met Ser Trp Val Arg Gln Thr Pro Asp Lys Arg Leu
    50                  55                  60

Glu Trp Ala Thr Ile Ser Asn Gly Gly Gly Glu Thr Tyr Tyr Glu Asp
65                  70                  75                  80

Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr
                85                  90                  95

Leu Tyr Leu Gln Met Asn Ser Leu Lys Ser Glu Asp Thr Ala Met Tyr
            100                 105                 110

Tyr Cys Ala Arg Glu Glu Tyr Glu Gly Thr Trp Tyr Glu Asp Val Trp
        115                 120                 125

Gly Ala Gly Thr Ala Lys Thr Thr Pro Pro Ser Val Tyr Pro Leu Ala
    130                 135                 140

Pro Gly Ser Ala Ala Gln Thr Asn Ser Met Val Thr Leu Gly Cys Leu
145                 150                 155                 160

Val Lys Gly Tyr Phe Pro Glu Pro Val Thr Val Thr Trp Asn Ser Gly
                165                 170                 175

Ser Leu Ser Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Asp
            180                 185                 190

Leu Tyr Thr Leu Ser Ser Ser Val Thr Val Pro Ser Ser Thr Trp Pro
        195                 200                 205

Ser Glu Thr Val Thr Cys Asn Val Ala His Pro Ala Ser Ser Thr Lys
    210                 215                 220

Val Asp Lys Lys Ile Val Pro Arg Asp Cys Gly Cys Lys Pro Cys Ile
225                 230                 235                 240

Cys Thr Val Pro Glu Val Ser Ser Val Phe Ile Phe Pro Pro Lys Pro
                245                 250                 255

Lys Asp Val Leu Thr Ile Thr Leu Thr Pro Lys Val Thr Cys Val Val
            260                 265                 270

Val Asp Ile Ser Lys Asp Asp Pro Glu Val Gln Phe Ser Trp Phe Val
        275                 280                 285
```

-continued

Asp Asp Val Glu Val His Thr Ala Gln Thr Gln Pro Arg Glu Glu Gln
290 295 300

Phe Asn Ser Thr Phe Arg Ser Val Ser Glu Leu Pro Ile Met His Gln
305 310 315 320

Asp Trp Leu Asn Gly Lys Glu Phe Lys Cys Arg Val Asn Ser Ala Ala
325 330 335

Phe Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly Arg Pro
340 345 350

Lys Ala Pro Gln Val Tyr Thr Ile Pro Pro Pro Lys Glu Gln Met Ala
355 360 365

Lys Asp Lys Val Ser Leu Thr Cys Met Ile Thr Asp Phe Phe Pro Glu
370 375 380

Asp Ile Thr Val Glu Trp Gln Trp Asn Gly Gln Pro Ala Glu Asn Tyr
385 390 395 400

Lys Asn Thr Gln Pro Ile Met Asp Thr Asp Gly Ser Tyr Phe Val Tyr
405 410 415

Ser Lys Leu Asn Val Gln Lys Ser Asn Trp Glu Ala Gly Asn Thr Phe
420 425 430

Thr Cys Ser Val Leu His Glu Gly Leu His Asn His His Thr Glu Lys
435 440 445

Ser Leu Ser His Ser Pro Gly Lys
450 455

<210> SEQ ID NO 15
<211> LENGTH: 236
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: light chain

<400> SEQUENCE: 15

Met Lys Leu Pro Val Arg Leu Leu Val Leu Met Phe Trp Ile Pro Ala
1 5 10 15

Ser Ser Ser Asp Val Leu Met Thr Gln Thr Pro Leu Ser Leu Pro Val
20 25 30

Ser Leu Gly Asp Gln Ala Ser Ile Ser Cys Thr Ser Ser Gln Ser Gly
35 40 45

Val His Ser Asn Gly Asn Thr Glu Leu Glu Trp Tyr Leu Gln Lys Pro
50 55 60

Gly Gln Pro Lys Leu Leu Ile Tyr Lys Val Ser Asn Arg Thr Ser Gly
65 70 75 80

Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
85 90 95

Lys Ile Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Tyr Cys Gly
100 105 110

Gln Gly Ser His Val Gly Arg Thr Phe Gly Gly Gly Thr Lys Leu Glu
115 120 125

Ile Lys Arg Ala Ala Ala Pro Thr Val Ser Ile Phe Pro Pro Ser Ser
130 135 140

Glu Gln Leu Thr Ser Gly Gly Ala Ser Val Val Cys Phe Leu Asn Asn
145 150 155 160

Phe Tyr Pro Lys Asp Ile Asn Val Lys Trp Lys Ile Asp Gly Ser Glu
165 170 175

-continued

```
Arg Gln Asn Gly Val Leu Asn Ser Trp Thr Asp Gln Asp Ser Lys Asp
            180                 185                 190

Ser Thr Tyr Ser Met Ser Ser Thr Leu Thr Leu Thr Lys Asp Glu Tyr
        195                 200                 205

Glu Arg His Asn Ser Tyr Thr Cys Glu Ala Thr His Lys Thr Ser Thr
    210                 215                 220

Ser Pro Ile Val Lys Ser Phe Asn Arg Asn Glu Cys
225                 230                 235


<210> SEQ ID NO 16
<211> LENGTH: 1392
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: encoding the heavy chain

<400> SEQUENCE: 16

atgaacttcg ggctcagctt gattttcctt gccctcattt taaaaggtgt ccagtgtgag     60

gtgcagctgg tggagtctgg gggagactta gtgaagcctg gagggtccct gaaactctcc    120

tgtgcagcct ctggattcac tttcagtagc ggcggcatgt cttgggttcg ccagactcca    180

gacaagaggc tggagtgggt cgcaaccatt agtaatggtg gtggtgagac ctactatgag    240

gacagtgtga aggggcgatt caccatctcc agagacaatg ccaagaacac cctgtaccta    300

caaatgaaca gtctgaagtc tgaggacaca gccatgtatt actgtgctag ggaggagtac    360

gagggtacgt ggtactgaga tgtctggggc gcagggacca cggtcaccgt ctcctcagcc    420

aaaacgacac ccccatctgt ctatccactg gcccctggat ctgctgccca aactaactcc    480

atggtgaccc tgggatgcct ggtcaagggc tatttccctg agccagtgac agtgacctgg    540

aactctggat ccctgtccag cggtgtgcac accttcccag ctgtcctgca gtctgacctc    600

tacactctga gcagctcagt gactgtcccc tccagcacct ggcccagcga gaccgtcacc    660

tgcaacgttg cccacccggc cagcagcacc aaggtggaca agaaaattgt gcccagggat    720

tgtggttgta agccttgcat atgtacagtc ccagaagtat catctgtctt catcttcccc    780

ccaaagccca aggatgtgct caccattact ctgactccta aggtcacgtg tgttgtggta    840

gacatcagca aggatgatcc cgaggtccag ttcagctggt ttgtagatga tgtggaggtg    900

cacacagctc agacgcaacc ccgggaggag cagttcaaca gcactttccg ctcagtcagt    960

gaacttccca tcatgcacca ggactggctc aatggcaagg agttcaaatg cagggtcaac   1020

agtgcagctt tccctgcccc catcgagaaa accatctcca aaaccaaagg cagaccgaag   1080

gctccacagg tgtacaccat tccacctccc aaggagcaga tggccaagga taaagtcagt   1140

ctgacctgca tgataacaga cttcttccct gaagacatta ctgtggagtg gcagtggaat   1200

gggcagccag cggagaacta caagaacact cagcccatca tggacacaga tggctcttac   1260

ttcgtctaca gcaagctcaa tgtgcagaag agcaactggg aggcaggaaa tactttcacc   1320

tgctctgtgt tacatgaggg cctgcacaac caccatactg agaagagcct ctcccactct   1380

cctggtaaat ga                                                        1392


<210> SEQ ID NO 17
<211> LENGTH: 717
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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```
<220> FEATURE:
<223> OTHER INFORMATION: encoding the light chain

<400> SEQUENCE: 17

atgaagttgc ctgttaggct gttggtgctg atgttctgga ttcctgcttc cagcagtgat     60

gttttgatga cccaaactcc actctccctg cctgtcagtc ttggagatca agcctccatc    120

tcttgcacat ctagtcagag cggagtacat agtaatggaa acaccgagtt agaatggtac    180

ctgcagaaac caggccagtc tccaaagctc ctgatctaca aagtttccaa ccgaacatct    240

ggggtcccag acaggttcag tggcagtgga tcagggacag atttcacact caagatcagc    300

agagtggagg ctgaggatct gggagtttat tactgcggcc aaggttcaca tgttggccgg    360

acgttcggtg gaggcaccaa gctggaaatc aaacgggctg atgctgcacc aactgtatcc    420

atcttcccac catccagtga gcagttaaca tctggaggtg cctcagtcgt gtgcttcttg    480

aacaacttct accccaaaga catcaatgtc aagtggaaga ttgatggcag tgaacgacaa    540

aatggcgtcc tgaacagttg gactgatcag gacagcaaag acagcaccta cagcatgagc    600

agcaccctca cgttgaccaa ggacgagtat gaacgacata acagctatac ctgtgaggcc    660

actcacaaga catcaacttc acccattgtc aagagcttca acaggaatga gtgttag       717
```

What is claimed is:

1. An antibody or antibody fragment thereof that specifically binds to voltage-gated sodium channel α subunit Nav1.7, wherein the antibody or antibody fragment specifically binds to an ion-conducting pore module of DIVS3 domain of the voltage-gated sodium channel α subunit.

2. The antibody or antibody fragment thereof according to claim **1**, characterized in that the antibody or antibody fragment comprises:

- heavy chain complementary determining regions HCDR1, HCDR2, HCDR3, the HCDR1 having the amino acid sequence as shown in SEQ ID NO. 1, the HCDR2 having the amino acid sequence as shown in SEQ ID NO. 2, and the HCDR3 having the amino acid sequence as shown in SEQ ID NO. 3; and
- light chain complementary decision regions LCDR1, LCDR2, LCDR3, the LCDR1 having the amino acid sequence as shown in SEQ ID NO. 4, the LCDR2 having the amino acid sequence as shown in SEQ ID NO. 5, and the LCDR3 having the amino acid sequence as shown in SEQ ID NO. 6.

3. The antibody or an antibody fragment thereof according to any one of claims **1**-**2**, characterized in that the antibody or antibody fragment comprises a heavy chain variable region as shown in SEQ ID NO.7 and/or a light chain variable region as shown in SEQ ID NO.8.

4. The antibody or antibody fragment thereof according to any one of claims **1**-**3**, characterized in that the antibody further comprises an antibody constant region.

5. The antibody or an antibody fragment thereof according to any one of claims **1**-**4**, the antibody or antibody fragment comprises a heavy chain as shown in SEQ ID NO. 14 and/or a light chain as shown in SEQ ID NO. 15.

6. The antibody or antibody fragment thereof according to claim **1**, characterized in that the antibody or antibody fragment is in a structural form selected from the group consisting of a full antibody, Fab, $F(ab')_2$, dsFv, scFv, a diabody, a minibody, a bispecific antibody, a multi-specific antibody, a chimeric antibody, and a CDR-grafted antibody.

7. The antibody or antibody fragment thereof according to claim **1**, characterized in that the antibody is a monoclonal antibody.

8. The antibody or antibody fragment thereof according to claim **7**, characterized in that the antibody is a humanized antibody.

9. A polypeptide that specifically binds to the antibody or antibody fragment according to claim **1**, wherein the polypeptide has an amino acid sequence as shown in SEQ ID NO.9.

10. An isolated nucleotide encoding the antibody or antibody fragment that specifically binds to voltage-gated sodium channel α subunit Nav1.7 according to any one of claims **1**-**8**.

11. An expression vector containing the nucleotide according to claim **10**.

12. A host cell containing the nucleotide according to claim **10** or the expression vector according to claim **11**.

13. A method of preparing the antibody or antibody fragment that specifically binds to the voltage-gated sodium channel α subunit Nav1.7 according to claim **1**, wherein the method comprising the steps of:

(a) culturing the host cell according to claim 12 under expression conditions so as to express the antibody or antibody fragment that specifically binds to the voltage-gated sodium channel α subunit Nav1.7;

(b) isolating and purifying the antibody or antibody fragment that specifically binds to the voltage-gated sodium channel α subunit Nav1.7.

14. A pharmaceutical composition, comprising the antibody or antibody fragment according to anyone of claims **1**-**8**.

15. Use of the antibody or antibody fragment according to claims **1**-**8** or of the pharmaceutical composition according to claim **14** for the manufacture of a medicament for the treatment of pain-related diseases.

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